

## 1 Making reinforcement learning work, and reporting it

The site page for this is a stub, but the instrument already exists in this project. `Literature/PaperReads/matrix_en` reads twenty one sources against twenty six fixed columns, and those columns are the checklist. This section states what the columns are for, what the methodology papers found, and where they disagree.

### 1.1 The twenty six columns

The lens is applied to every source in the same order, so answers are comparable across papers. Grouped by what they check:

**Statistical practice.** Seeds. Variance reporting. Random source control. Statistical tests. Diverged-run handling.

**Hyperparameters and convergence.** Hyperparameter reporting. Hyperparameter sensitivity. Convergence demonstration.

**Comparison.** Calibration baselines. Baseline tuning. Implementation matching. Environment selection. Comparison style.

**The problem as posed.** Episode length. Discount factor. Reward function. Observation construction. Action space.

**Learning machinery.** Exploration. Credit assignment. Replay buffer. Offline learning support.

**Evidence.** Generalization. Ablation discipline. Sample complexity against compute. Action latency budget.

Two properties make it work. The columns are fixed before reading, so a paper cannot set its own bar. And *not reported* is a legal answer, so silence is recorded rather than filled in. Several entries in the matrix read “Not reported” or “Not discussed,” which is a finding about the field.

### 1.2 Seeds, and how many is enough

Four sources answer this and they do not agree on a number. All four agree that five is too few.

**Henderson and colleagues**, AAAI 2018, ran five trials per evaluation with preset seeds shared across conditions. Their demonstration is the useful part: they split ten trials into two groups of five, ran the same algorithm with the same hyperparameters in both, and found the two groups statistically different at  $t = -9.0916$ ,  $p = 0.0016$ . Two identical configurations produced a significant difference through seed choice alone.

They also argue against a common practice: “it is not uncommon for the top-N trials to be selected from among several trials ... Our experiment with random seeds shows that this can be potentially misleading.”

**Colas and colleagues** treat the question directly. With  $N = 5$  the bootstrap test produced a false positive rate near 10 percent against a nominal 5 percent. Welch’s t-test came in slightly above 5 percent. Their recommendations are concrete: use a larger sample than the power analysis prescribes, prefer Welch’s t-test to the bootstrap test, apply a Bonferroni correction for multiple comparisons, and set the significance level below 0.05 so the empirical type one error stays under

0.05.

**Agarwal and colleagues**, NeurIPS 2021 Outstanding Paper, ran 100 runs per algorithm and game across five algorithms and twenty six games, then subsampled at 3, 5, 10, 25, 50, and 100 to study how the conclusion moves. Their assessment of what the field would need: the number is “closer to 50–100 runs in Atari 100k – far too many to be computationally feasible for most research projects.”

They also identify a bias that matters at small sample sizes: “the sample median is a biased estimator of the true median ... in the few-run regime, we find that this bias can dominate the comparison between algorithms.”

**Engstrom and colleagues**, ICLR 2020, trained “at least 80 agents for each estimate (more for some high-variance cases),” with 95 percent confidence intervals from a thousand-sample bootstrap, and added agents where variance was high rather than reporting a wider interval.

The systematization of pitfalls in reinforcement learning for cybersecurity sets the bar at twenty runs with 95 percent confidence intervals, and records that a large share of surveyed papers fail at least one reporting criterion.

### 1.3 Where the sources disagree

One disagreement is unresolved and should stay that way rather than being averaged.

Colas recommends a significance test, names Welch’s t-test as the preferred one, and gives the correction and the significance level to use with it.

Agarwal recommends against significance tests: “we recommend using confidence intervals ... but avoid statistical significance tests (e.g., p-value < 0.05) because of their dichotomous nature ... and common misinterpretations.” In their place: stratified bootstrap confidence intervals, performance profiles, the interquartile mean as a robust alternative to the median, the optimality gap, and the average probability of improvement.

The empirical design source sides with intervals, recommending paired difference confidence intervals over hypothesis tests.

Both positions are defensible and they lead to different tables. Choose one, state which, and do not mix them.

### 1.4 What this project decided

The position on record in `reading_insights.md`, arrived at after the disagreement above was surfaced: five seeds, plus paired bootstrap confidence intervals, plus per-seed worst-case disclosure.

That is below the twenty run bar and far below Agarwal’s fifty to one hundred. The disclosure is what makes it honest. Raw per-seed episode losses ship with the artifact bundle so an independent bootstrap can be computed, which is stated in the paper’s evaluation setup.

Two facts about the current paper follow from the columns and are worth having in view. Five seeds is stated. Statistical tests are not reported, and the ordering claim rests on the gap exceeding one standard deviation in every cell.

## 1.5 Baselines, and the three ways a comparison fails

The columns separate three failures that are usually collapsed.

**Calibration baselines** ask whether a trivial policy would have done as well. A random policy and a do-nothing policy bracket the range and cost almost nothing to run. The current paper has three such references, Sleep, Random, and an expert-inspired heuristic, and the heuristic is the one that carries the argument because it encodes the standard playbook as a fixed policy.

**Baseline tuning** asks whether the comparison method was given the same care as the proposed one. The empirical design source names untuned baselines and designer bias as endemic. Henderson demonstrates the size of the effect by varying one hyperparameter at a time across four algorithms.

**Implementation matching** asks whether the two methods came from the same code. Henderson compared three codebases for one algorithm and three for another, and found the codebase itself changed the result. Engstrom took the stronger route, running everything from one codebase and listing every parameter, and states that it acknowledges its own bug.

The distinction that Engstrom isolates is worth carrying: code-level optimizations can account for a difference attributed to the algorithm. Their design separates the step method from the code-level options and runs all combinations, which is the only way to tell the two apart.

## 1.6 The nine challenges, and which ones a project addresses

Dulac-Arnold and colleagues name nine challenges of real-world reinforcement learning, each restated as a perturbation of the underlying decision process, each isolated, then recombined. The list is a yardstick a reader already knows.

The operative move for a paper is calibration rather than coverage. Name which challenges the work attacks, which it inherits, and which it puts out of scope. For this project the notes record the match as learning on a live system from limited samples, partial observability with a moving adversary, and the multiple objective tension between containment and keeping users working. Out of scope: low latency constraints, explainability, learning from logged data, and hard safety constraints.

Stating the out-of-scope list is the part that gets skipped and the part that prevents a reviewer from reading a partial result as a universal claim.

## 1.7 Two rules that apply before any of the above

**Do not form the hypothesis after seeing the results.** The empirical design source names this directly, and its recommended workflow puts the scientific question, the hyperparameter sets, and the seed budget in writing before the runs start. The look back step is honest re-examination rather than a narrative fitted to what came out.

**Convergence goes in the body.** A learning curve in an appendix is a claim the reader is asked to take on trust. Henderson also warns that a curve can hide a local optimum, so the curve alone is not sufficient evidence that the policy is doing what its score suggests.

## 1.8 Sources

- `Literature/PaperReads/matrix_empirical_rl_lens.md`, this project’s own instrument. Twenty six columns applied to twenty one sources. All quotations below reach this file through that matrix and its accompanying quality reports rather than through a fresh read of each paper, which is recorded here so the access level is not overstated.
- P. Henderson, R. Islam, P. Bachman, J. Pineau, D. Precup, D. Meger, *Deep Reinforcement Learning That Matters*, AAAI 2018.
- C. Colas, O. Sigaud, P.-Y. Oudeyer, *How Many Random Seeds? Statistical Power Analysis in Deep Reinforcement Learning Experiments*, arXiv 2018, with a later journal version.
- R. Agarwal, M. Schwarzer, P. S. Castro, A. Courville, M. G. Bellemare, *Deep Reinforcement Learning at the Edge of the Statistical Precipice*, NeurIPS 2021, Outstanding Paper. Companion library `rliable` at [github.com/google-research/rliable](https://github.com/google-research/rliable).
- L. Engstrom, A. Ilyas, S. Santurkar, D. Tsipras, F. Janoos, L. Rudolph, A. Madry, *Implementation Matters in Deep Policy Gradients*, ICLR 2020.
- G. Dulac-Arnold and colleagues, *Challenges of Real-World Reinforcement Learning*, Machine Learning Journal 2021, arXiv 2003.11881.
- The systematization of pitfalls in deep reinforcement learning for cybersecurity, and the empirical design source, both read for this project and recorded in **PaperReads**.

The grouping of the twenty six columns into six kinds is not in the matrix. The matrix lists them in order without grouping.